

WESTERLY, RI, USA

WMO: 722151

Lat: 41.350N

Lon: 71.799W

Elev: 25

StdP: 101.03

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 14794

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.3	-22.7	0.5	-11.4	-20.2	0.6	-7.8	10.6	1.5	9.3	2.7	3.0	310	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	29.8	22.7	28.4	22.1	27.2	21.5	24.6	27.7	23.8	26.5	23.0	25.4	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	26.3	22.9	17.6	25.2	22.2	16.9	24.6	74.5	27.7	71.1	26.6	68.1	25.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	6.9	-17.4	32.7	3.6	1.6	-20.0	33.8	-22.1	34.8	-24.1	35.7	-26.7	36.9		
			-18.2	25.6	3.3	1.0	-20.5	26.3	-22.4	26.9	-24.3	27.5	-26.7	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-0.7	-0.1	3.3	8.8	14.0	18.8	22.6	21.9	18.4	12.7	7.0	2.6
	DBStd	9.03	5.35	4.78	4.41	3.54	3.48	3.09	2.50	2.68	3.38	4.21	4.34	4.81
	HDD10.0	1257	332	284	211	63	5	0	0	0	0	21	108	233
	HDD18.3	3082	590	517	466	286	141	31	1	3	41	181	339	487
	CDD10.0	1563	1	1	3	27	129	264	391	368	251	104	19	4
	CDD18.3	347	0	0	0	1	7	45	134	113	43	6	0	0
	CDH23.3	1948	0	0	0	6	33	217	821	698	163	10	0	0
	CDH26.7	373	0	0	0	1	3	35	183	131	18	2	0	0
Wind	WSAvg	3.0	3.3	3.3	3.5	3.4	3.0	2.8	2.6	2.4	2.6	2.9	3.0	3.1
Precipitation	PrecAvg	1136	88	79	113	101	86	85	82	102	106	99	87	108
	PrecMax	1395	153	141	360	195	195	281	288	209	219	330	142	189
	PrecMin	876	40	40	12	32	27	0	34	39	24	10	21	23
	PrecStd	147	31	30	72	47	37	65	49	48	49	66	35	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	12.4	17.3	23.2	26.8	30.0	32.3	31.3	28.6	24.5	18.6	15.5
		MCWB	11.3	9.0	11.4	14.3	19.0	21.7	24.2	23.8	22.6	19.4	15.3	13.4
	2%	DB	11.0	10.4	13.6	18.8	23.8	27.6	30.1	29.4	26.6	22.1	17.2	13.8
		MCWB	10.1	8.4	8.6	11.2	17.5	20.7	22.9	22.9	20.7	19.3	15.0	12.4
	5%	DB	9.0	8.9	11.5	16.3	21.8	25.9	28.7	28.1	25.1	20.9	15.9	12.3
		MCWB	7.4	6.8	8.2	10.8	16.1	19.6	22.4	22.2	20.5	18.3	13.5	11.0
	10%	DB	7.5	7.2	9.8	14.5	20.0	24.2	27.4	26.9	23.9	19.5	14.5	10.6
		MCWB	5.9	5.0	6.8	10.0	15.1	18.7	21.8	21.7	20.2	16.7	12.5	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	11.1	13.0	16.0	20.6	23.3	25.8	25.7	24.3	21.7	17.6	14.6
		MCDB	12.3	11.6	14.6	19.4	24.8	27.9	30.4	29.1	26.3	22.6	18.1	15.1
	2%	WB	10.3	9.0	10.6	13.9	18.9	21.9	24.6	24.9	23.1	20.2	15.7	12.7
		MCDB	10.7	9.7	12.2	16.5	22.4	25.8	28.2	27.6	24.9	21.5	16.6	13.4
	5%	WB	7.9	7.3	8.9	12.6	17.6	21.0	23.8	24.0	22.1	18.9	14.3	11.3
		MCDB	8.7	8.7	10.7	14.7	20.5	24.0	26.8	26.5	23.8	20.4	15.3	12.1
	10%	WB	5.9	5.3	7.4	11.1	16.3	20.1	23.0	23.1	21.2	17.6	12.9	9.1
		MCDB	7.3	6.9	9.2	13.1	18.8	22.6	25.5	25.2	23.0	19.0	14.4	10.4
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.5	8.3	9.0	8.6	8.3	8.2	8.7	9.2	9.2	9.0	7.9
		MCDBR	9.4	9.4	10.1	11.9	11.0	10.4	9.7	9.4	9.5	9.4	9.3	8.9
		MCWBR	9.4	8.0	7.0	6.4	5.7	5.0	4.2	4.5	5.7	6.7	8.3	9.1
	5% WB	MCDBR	9.4	9.7	8.8	9.0	9.1	8.6	7.9	7.2	7.2	7.6	8.0	8.8
		MCWBR	9.9	9.2	7.2	6.3	5.5	4.9	3.9	4.0	5.0	6.3	8.3	9.4
Clear-Sky Solar Irradiance	taub	0.307	0.315	0.338	0.375	0.423	0.464	0.488	0.450	0.379	0.345	0.317	0.304	
	taud	2.495	2.471	2.449	2.378	2.276	2.202	2.145	2.243	2.414	2.516	2.553	2.553	
	Ebn at Noon	850	898	913	896	856	819	794	814	856	852	835	827	
	Edh at Noon	74	88	101	116	132	143	150	132	104	84	70	65	
All-Sky Solar Radiation	RadAvg	1.87	2.75	3.77	4.89	5.42	5.93	6.05	5.43	4.44	3.00	2.05	1.56	
	RadStd	0.15	0.27	0.28	0.52	0.53	0.59	0.32	0.30	0.34	0.28	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.64	N/A	N/A	+1.35	N/A	N/A	N/A	N/A	+196	+102
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	+55

Nomenclature: See separate page